

Code:BCS505**Course: Full Stack Development****Credits: 3****L:T:P - 3:0:0****SEE: 50****CIE: 50****SEE Hours: 3 Max. Marks: 100**

Prerequisites if any	Computer networks
Learning objectives	1. Create web page using HTML & CSS 2. Develop familiarity with the JavaScript language, realize concepts commonly used in dynamic language programming, such as introspection, higher-order functions, closures, familiar with common libraries and tools that are used in web application development. 3. To Create React Components, lifecycle of components, rendering list and portal and perform some simple tests, and error handling. 4. To create Node.js modules, events and database access and interact with databases using MongoDB .

Course Outcomes:*On the successful completion of the course, the student will be able to*

COs	Course Outcomes	Bloom's level
CO1	Prepare web pages using HTML and CSS.	Apply
CO2	Apply core constructs and event handling mechanisms of JavaScript to develop webPages.	Apply
CO3	Build interactive web applications using ReactJS	Analyze
CO4	Develop interactive web applications using Node.js and Express.js.	Apply
CO5	Demonstrate the creation of dynamic web pages and connecting to MYSQL/MongoDB	Apply

Mapping with POs and PSOs:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2
CO1	3	2	3	2	2	1	-	1	-	-	1	2		3	3
CO2	3	2	3	2	2	1	-	1	-	-	1	2		3	3
CO3	3	2	3	2	3	1	-	1	-	-	1	2		3	3
CO4	3	2	3	2	3	1	-	1	-	-	1	2		3	3
CO5	3	2	3	2	3	1	-	1	-	-	1	2		3	3

Mapping Strength: Strong– 3 Medium – 2 Low – 1

Course Structures

Nos.	Modules	No. of Lecture Hours	No. of Tutorial Hours	No. of Practical Hours
Modules -1:HTML &CSS				
1.1	Introduction to HTML, Basic syntax, Standard HTML Document Structure, Images,	1	-	-
1.2	Paragraphs, Line Breaks, Heading Tags, Font Styles, Sizes, Color, Character Entity	1	-	-
1.3	links, Lists, tables, form tags, Font properties, text decoration, List property,	2	-	-
1.4	CSS Colors and background, CSS Box Model, CSS Margins, Padding, Borders. Background images, span and div tags, Grid and flex.	2	-	-
Module 2: JavaScript				
2.1	Introduction to JavaScript, General syntactic characteristics	1	-	-
2.2	Primitives, operations, expressions	1	-	-
2.3	Screen output and keyboard input, Control statements,	1	-	-
2.4	Object creation &modification, arrays, array methods, Functions,	2	-	-
2.5	JavaScript and XHTML Documents: The JavaScript execution environment, TheDocument Object Model, Element access in JavaScript, Events and event handling,	2	-	-
2.6	Handling events from the Body elements, Button elements, Text box and Password elements.	2	-	-
Module – 3: ReactJS				
3.1	Introduction, Templating using JSX	1	-	-
3.2	Components, State, Props, Hooks	2	-	-
3.3	Lifecycle of Components	2	-	-
3.4	Rendering List and Portals	2	-	-
3.5	Error Handling, Routers	2	-	-
Module4: NodeJS				
4.1	Node.js overview, Node.js – basics and setup.	1	-	-
4.2	Node.js console, Node.js command utilities	1	-	-

4.3	Node.js modules, Node.js concepts	2	-	-
4.4	Node.js events, Node.js database access.	2		-
4.5	Introduction to Express.js • Creating routes (GET, POST, PUT, DELETE) • Handling request and response objects • Middleware in Express (built-in, custom, third-party)	2		
Module 5: MongoDB				
5.1	MongoDB basics: Documents, collections, database query language, mongo shell.	2	-	-
5.2	MongoDB CRUD operations (create, read, projection, update, delete aggregate)	2	-	-
5.3	reading from MongoDB, writing from MongoDB	2	-	-
5.4	MongoDB with PHP, MongoDB with NodeJS	2	-	-
	Total No. of Lecture Hours	40		-
	Total No. of Tutorial Hour		00	-
	Total No. of Practical Hours			00

Textbook:

1. Programming the World Wide Web, Robert W. Sebesta, 4th Edition, Pearson Education, 2008.
2. Full-Stack React Projects: Learn MERN Stack Development by Building Modern Web Apps Using MongoDB, Express, React, and Node.js, 2nd Edition.

Reference Book:

1. Open-Source Web Development with Lamp, James Lee and Brent Ware, Pearson Education, 2009.
2. Internet and World Wide Web: How to Program -Harvey M. Deitel, Paul J. Deitel, 4th edition, Pearson, 2009.
3. The Web Programming Building Internet Applications- Chris Bates, 3rd Edition, Wiley India, 2006

Online Resources:

1. MongoDB Notes for Professionals book Tutorial for Beginners in PDF by GoalKicker.com
2. <https://docs.google.com/viewer?url=https://www.computer-pdf.com/pdf/0840-mongodb-notes-for-professionals-book.pdf>
3. <https://www.javatpoint.com/reactjs-tutorial>